EMC – Practice Questions

1 EMC First Midterm --- Practice Questions

1.1 Chapter 1: Introduction to EMC

1.1.1 Conceptual Questions

Q1.1 *(chap1.pdf, p.4)* Define electromagnetic compatibility. List the three criteria that must be simultaneously satisfied for a system to be EMC-compliant.

Q1.2 *(chap1.pdf, p.5)* True or False: If an unintended EM signal is picked up by a radio receiver, this constitutes electromagnetic interference. Explain your answer.

Q1.3 *(chap1.pdf, p.6)* List the three general approaches to preventing EMI in order of cost-effectiveness. Why is the first approach usually the most effective?

Q1.4 *(chap1.pdf, p.7)* Name and briefly describe the four basic EMC subproblems. Which two involve the power cable?

Q1.5 *(chap1.pdf, p.9)* What is ESD, and approximately how large can the voltage become? Why is it a concern even though it is not regulated in the same way as FCC emissions?

Q1.6 *(chap1.pdf, p.15)* A circuit element has physical length 2 cm. Under what conditions can you safely use lumped-circuit analysis (KVL, KCL) to model it? Express your answer in terms of wavelength.

1.1.2 Short Calculation

Q1.7 *(chap1.pdf, pp.20--21)* A signal at 3 GHz propagates in a PCB dielectric with $\varepsilon_r = 4$ (assume $\mu_r = 1$). (a) Find the phase velocity v. (b) Find the wavelength λ. (c) What is the maximum trace length that allows lumped-circuit modelling?

Q1.8 *(chap1.pdf, pp.25--26)* Express the following gains in decibels: (a) Power gain of 500 (b) Voltage gain of 30 (c) A factor of 1/4 in power (d) A factor of 1/4 in voltage

Q1.9 *(chap1.pdf, p.28)* A -5 dBm signal passes through a cable with 4 dB of loss, then through an amplifier with 35 dB gain, then through another cable with 2 dB of loss. What is the output power in dBm?

Q1.10 *(chap1.pdf, p.30)* A 50-m coaxial cable has a measured power loss of 0.5 dB. Find the attenuation constant α in Np/m.

Q1.11 *(chap1.pdf, p.30)* A spectrum analyser shows a signal level of 80 dBμV (RMS). The measurement is performed at $R = 50\,\Omega$. What is the signal power in dBm?

Q1.12 *(chap1.pdf, p.17)* A 3 GHz sinusoidal signal propagates along a 4.5 cm trace on FR-4 ($\varepsilon_r = 4$). What is the phase difference between the input and output of the trace?

1.2 Chapter 2: Regulatory Requirements

1.2.1 Conceptual Questions

Q2.1 *(chap2.pdf, p.6)* What is the difference between FCC Class A and Class B devices? Which class has stricter limits, and why?

Q2.2 *(chap2.pdf, p.7)* State the frequency ranges for (a) conducted emission testing and (b) radiated emission testing under FCC regulations. What instrument is used for conducted emission measurements?

Q2.3 *(chap2.pdf, p.13)* FCC Class A radiated emission limits are measured at 10 m and Class B at 3 m. After normalising both to 3 m, how many dB less stringent are the Class A limits compared to Class B? Show the calculation.

Q2.4 *(chap2.pdf, p.16)* In what important way did the European EMC Directive (CISPR 22) go further than FCC Part 15?

Q2.5 *(chap2.pdf, pp.25--27)* Describe the construction of a semianechoic chamber (SAC). What detector type is used for FCC measurements? For MIL-STD-461E measurements?

Q2.6 *(chap2.pdf, pp.28--32)* Explain the two purposes served by the LISN in conducted emission testing. Why must its impedance be constant with frequency?

Q2.7 *(chap2.pdf, p.41)* True or False: The best time to address EMC in a product design is during the board layout phase. Explain the preferred approach.

Q2.8 *(chap2.pdf, p.17)* How do MIL-STD-461E requirements compare to FCC/CISPR 22 in terms of frequency range and flexibility?

1.3 Chapter 3: Signal Spectra

1.3.1 Conceptual Questions

Q3.1 *(chap3.pdf, p.3)* What property of a digital clock signal most strongly determines its high-frequency spectral content, and why does it matter for EMC?

Q3.2 *(chap3.pdf, p.4)* Why must a data stream in a digital communication system be nondeterministic (random)?

Q3.3 *(chap3.pdf, p.9)* A signal has complex Fourier coefficients c_n. How do you obtain the one-sided (positive-frequency) spectrum from these? What happens to the DC component c_0?

Q3.4 *(chap3.pdf, p.14)* A symmetric square wave has a 50% duty cycle. What can be said about its even harmonics? Explain physically why they vanish.

Q3.5 *(chap3.pdf, pp.33--35)* Sketch the Bode-plot spectral envelope of a trapezoidal clock waveform. Label the two breakpoint frequencies, the DC level, and the slopes in each region. What design change has the greatest effect on reducing high-frequency emissions?

Q3.6 *(chap3.pdf, p.37)* A digital clock has $\tau_r = 2$ ns. Estimate its signal bandwidth. Where does the first spectral null occur in the true spectrum?

Q3.7 *(chap3.pdf, pp.40--42)* What causes ringing in digital signals on PCBs? How does ringing affect the frequency spectrum? List two circuit-level remedies.

Q3.8 *(chap3.pdf, p.46)* What is a quasi-peak detector, and why is it required for regulatory measurements instead of a simple peak detector?

1.3.2 Short Calculation

Q3.9 *(chap3.pdf, pp.11--12)* A periodic rectangular pulse train has amplitude $A = 5$ V, period $T = 20$ ns, pulse width $\tau = 5$ ns. (a) Find the duty cycle D. (b) Find the DC component c_0. (c) Find $|c_3|$ (the magnitude of the 3rd harmonic coefficient).

Q3.10 *(chap3.pdf, pp.33--34)* For the clock in Q3.9, add rise time $\tau_r = 1$ ns. Find: (a) The DC level of the spectral envelope ($2AD$). (b) The first breakpoint f_1. (c) The second breakpoint f_2. (d) The bandwidth BW.

Q3.11 *(chap3.pdf, p.36)* Using the spectral envelope from Q3.10, estimate the spectral level (in V) at $f = 2$ GHz relative to the DC level. Express as a dB ratio.

Q3.12 *(chap3.pdf, pp.38--39)* Compare two clock signals both with $A = 3.3$ V, $T = 10$ ns, $\tau_r = 0.5$ ns: Clock 1: $\tau = 5$ ns ($D = 0.5$) Clock 2: $\tau = 1$ ns ($D = 0.1$)

(a) What is the DC spectral level for each?
(b) For each, at what frequency does the flat region end?
(c) What happens to the high-frequency ($f > f_2$) content when we switch from Clock 1 to Clock 2?

Q3.13 *(chap3.pdf, p.20)* A periodic signal $x(t)$ has Fourier coefficients c_n. Signal $y(t) = x(t - 3\text{ ns})$ is a delayed version of $x(t)$ with period $T = 10$ ns. (a) Write the Fourier coefficients of $y(t)$ in terms of c_n. (b) Does the time delay affect the magnitude of any harmonic? Explain.

1.4 Chapter 4: Transmission Lines

1.4.1 Conceptual Questions

Q4.1 *(chap4.pdf, p.3)* What are the two main effects that make a conductor at high frequencies behave as a transmission line rather than a simple wire?

Q4.2 *(chap4.pdf, p.7)* What physical phenomenon causes the per-unit-length resistance of a transmission line to increase significantly at GHz frequencies?

Q4.3 *(chap4.pdf, pp.4--5)* Why is a coaxial cable non-dispersive while a microstrip is dispersive? Relate your answer to the medium surrounding the conductors.

Q4.4 *(chap4.pdf, p.12)* For a TEM-mode transmission line in a homogeneous medium, write the relationship between the per-unit-length inductance l, capacitance c, and the medium parameters μ and ε. What does this imply about the phase velocity?

Q4.5 *(chap4.pdf, p.35)* The current reflection coefficient is the **negative** of the voltage reflection coefficient. Explain physically why this makes sense (consider what happens to current direction upon reflection from an open circuit vs. a short circuit).

Q4.6 *(chap4.pdf, pp.54--55)* Compare series matching and parallel matching. For each: (a) state the design condition, (b) state whether the load sees the full source voltage, (c) state whether the matching resistor consumes power.

Q4.7 *(chap4.pdf, p.59)* State both forms of the criterion for when a transmission line can be treated as a lumped element. Express one criterion in terms of rise time and one in terms of electrical length.

Q4.8 *(chap4.pdf, p.87)* Explain why dielectric loss increases directly with frequency while skin-effect resistance increases only with $\sqrt{f}$. What does this mean for which loss mechanism dominates at very high frequencies?

Q4.9 *(chap4.pdf, pp.92--94)* For a typical PCB above what frequency is the low-loss approximation valid? Above what frequency does β dispersion disappear? Does α dispersion ever disappear?

1.4.2 Short Calculation

Q4.10 *(chap4.pdf, pp.18--20)* Two 1 mm radius wires are separated by 20 mm centre-to-centre. Find the approximate per-unit-length inductance l and capacitance c in air ($\varepsilon_r = \mu_r = 1$). (Hint: use the approximate formulas valid for $d \gg a$.)

Q4.11 *(chap4.pdf, pp.25--26)* A coaxial cable has inner radius $a = 1$ mm and outer radius $b = 5$ mm. The dielectric has $\varepsilon_r = 2.25$. Find: (a) l [H/m], (b) c [F/m], (c) Z_C [Ω], (d) v [m/s].

Q4.12 *(chap4.pdf, pp.34--37)* A lossless line has $Z_C = 75\,\Omega$, $v = 2 \times 10^8$ m/s, length $L = 30$ cm. The source is $V_S = 10$ V step with $Z_S = 75\,\Omega$. The load is $Z_L = 300\,\Omega$. (a) Find T_D. (b) Find V^+_{init}. (c) Find Γ_L and Γ_S. (d) Find the load voltage at $t = T_D$, $t = 3T_D$, and $t = \infty$.

Q4.13 *(chap4.pdf, pp.59--63)* A line has $T_D = 0.3$ ns. A signal with $\tau_r = 2$ ns drives the line. Does the line distort the signal significantly? Justify with the criterion.

Q4.14 *(chap4.pdf, p.77)* A line has $Z_C = 50\,\Omega$ and is terminated with $Z_L = 150 + j50\,\Omega$. (a) Find $|\Gamma_L|$. (b) Find the VSWR. (c) Find the fraction of incident power delivered to the load.

Q4.15 *(chap4.pdf, pp.82--83)* A 1 mm radius copper wire ($\sigma = 5.8 \times 10^7$ S/m) carries current at 100 MHz. (a) Find the skin depth δ. (b) Is the high-frequency resistance formula applicable (i.e. is $r_w \gg \delta$)? (c) Find the per-unit-length resistance r_{hf}.

1.5 Answers

1.5.1 Chapter 1 Answers

A1.1 Three criteria *(chap1.pdf, p.4)*: (1) does not cause interference with others; (2) is not susceptible to others; (3) does not interfere with itself.

A1.2 False. *(chap1.pdf, p.5)* Receiving an unintended signal constitutes interference only if the receiver **behaves in an undesired manner**. Mere coupling is not interference if the system still works correctly.

A1.3 *(chap1.pdf, p.6)* (1) Suppress source, (2) attenuate coupling path, (3) harden receptor. The source fix works because it reduces the problem at the root before it propagates.

A1.4 *(chap1.pdf, p.7)* Radiated emissions, radiated susceptibility, conducted emissions, conducted susceptibility. Conducted emissions and conducted susceptibility involve the power cable.

A1.5 *(chap1.pdf, p.9)* ESD: static discharge; can approach 25 kV. Concern in design because it can damage semiconductors or cause malfunction even if not government-regulated like emissions.

A1.6 *(chap1.pdf, p.15)* Safe to use lumped-circuit theory when $L < \lambda/10$.

A1.7 $v = 3 \times 10^8/2 = 1.5 \times 10^8$ m/s; $\lambda = 1.5 \times 10^8/(3 \times 10^9) = 0.05$ m $= 5$ cm; max lumped length $= 5$ cm$/10 = 0.5$ cm. *(chap1.pdf, pp.20--21)*

A1.8 (a) $10 \log 500 = 27.0$ dB; (b) $20 \log 30 = 29.5$ dB; (c) $10 \log(1/4) = -6.0$ dB; (d) $20 \log(1/4) = -12.0$ dB. *(chap1.pdf, pp.25--26)*

A1.9 $-5 - 4 + 35 - 2 = +24$ dBm. *(chap1.pdf, p.28)*

A1.10 $P_L = 0.5$ dB/50 m $= 0.01$ dB/m; $\alpha = 0.01/8.686 = 1.15 \times 10^{-3}$ Np/m. *(chap1.pdf, p.30)*

A1.11 dBm $= 80 - 107 = -27$ dBm. *(chap1.pdf, p.30)*

A1.12 $v = v_0/2 = 1.5 \times 10^8$ m/s; $\beta = 2\pi \times 3 \times 10^9/(1.5 \times 10^8) = 125.7$ rad/m; $\Delta\phi = 125.7 \times 0.045 = 5.66$ rad $\approx 324°$. *(chap1.pdf, p.17)*

1.5.2 Chapter 2 Answers

A2.1 *(chap2.pdf, p.6)* Class A: commercial/industrial; Class B: residential. Class B is stricter because residential users lack expertise/resources to deal with interference.

A2.2 *(chap2.pdf, p.7)* (a) 150 kHz -- 30 MHz (LISN); (b) 30 MHz -- 40 GHz (OATS or SAC).

A2.3 $20\log(10/3) = 10.46$ dB; Class A is ≈ 10 dB less stringent. *(chap2.pdf, p.13)*

A2.4 *(chap2.pdf, p.16)* CISPR 22/EU directive was first to require susceptibility (immunity) testing for commercial products --- FCC only requires emission limits.

A2.5 *(chap2.pdf, pp.25--27)* Top+sides: absorbers; floor: ground plane. FCC: quasi-peak; MIL: peak.

A2.6 *(chap2.pdf, pp.28--32)* (1) Present constant 50 Ω to EUT; (2) reject 60 Hz, pass RF noise to receiver. Constant impedance is required so the measurement gives unambiguous, repeatable results.

A2.7 False. *(chap2.pdf, p.41)* The preferred approach is "paper work before layout" --- address EMC during design/planning, not after the board is fabricated.

A2.8 *(chap2.pdf, p.17)* MIL-STD-461E spans much larger frequency ranges and is more complex, but requirements can be waived/tailored --- unlike FCC/CISPR which are fixed.

1.5.3 Chapter 3 Answers

A3.1 *(chap3.pdf, p.3)* Rise/fall time. Faster edges produce higher-frequency spectral content, which radiates more efficiently from PCB traces acting as antennas.

A3.2 *(chap3.pdf, p.4)* If the data were deterministic, no information would be conveyed --- the receiver could predict it in advance.

A3.3 *(chap3.pdf, p.9)* $c_n^+ = 2|c_n|$ for $n > 0$; c_0 is unchanged (already real).

A3.4 *(chap3.pdf, p.14)* Even harmonics vanish. At $D = 1/2$: $\sin(n\pi/2) = 0$ for even $n \to c_n = 0$. Physically, the +ve and -ve pulse contributions cancel for even harmonics.

A3.5 *(chap3.pdf, pp.33--34)* Flat at $2AD$ for $f < 1/(\pi\tau)$; -20 dB/decade for $1/(\pi\tau) < f < 1/(\pi\tau_r)$; -40 dB/decade above $1/(\pi\tau_r)$. Increasing rise time is most effective.

A3.6 $BW \approx 1/(2 \times 10^{-9}) = 500$ MHz; first null at $1/\tau_r = 500$ MHz. *(chap3.pdf, p.37)*

A3.7 *(chap3.pdf, pp.40--42)* PCB trace inductance and capacitance. Ringing adds emission peaks at the ringing frequency. Remedies: series resistor at gate output, ferrite beads.

A3.8 *(chap3.pdf, p.46)* QP responds more quickly to rising signals than falling; gives a weighted average that accounts for annoyance level. Required by FCC and CISPR.

A3.9 (a) $D = 5/20 = 0.25$; (b) $c_0 = AD = 5 \times 0.25 = 1.25$ V; (c) $c_3 = 5 \times 0.25 \times |\sin(3\pi \times 0.25)/(3\pi \times 0.25)| = 1.25 \times |\sin(0.75\pi)/(0.75\pi)| = 1.25 \times (0.7071/2.356) = 0.375$ V. *(chap3.pdf, pp.11--12)*

A3.10 (a) $2AD = 2 \times 5 \times 0.25 = 2.5$ V; (b) $f_1 = 1/(\pi \times 5 \times 10^{-9}) = 63.7$ MHz; (c) $f_2 = 1/(\pi \times 10^{-9}) = 318$ MHz; (d) $BW = 1/\tau_r = 1$ GHz. *(chap3.pdf, pp.33--34, 37)*

A3.11 2 GHz is in the -40 dB/decade region (above $f_2 = 318$ MHz). From f_2 to 2 GHz: $\log(2000/318) = 0.799$ decades $\to 40 \times 0.799 = 32$ dB extra drop beyond the -20 dB/decade drop from f_1 to f_2. Drop f_1 to f_2: $20\log(318/63.7) = 20 \times 0.699 = 14$ dB. Total drop at 2 GHz: $14 + 32 = 46$ dB below DC level. *(chap3.pdf, pp.33--36)*

A3.12 (a) Clock 1: $2 \times 3.3 \times 0.5 = 3.3$ V; Clock 2: $2 \times 3.3 \times 0.1 = 0.66$ V (14 dB lower). (b) Clock 1: $f_1 = 1/(\pi \times 5 \text{ ns}) = 63.7$ MHz; Clock 2: $f_1 = 1/(\pi \times 1 \text{ ns}) = 318$ MHz. (c) No change --- both have same $\tau_r = 0.5$ ns $\to$ same $f_2 = 637$ MHz $\to$ same -40 dB/decade high-frequency slope. *(chap3.pdf, pp.38--39)*

A3.13 (a) $c_n^y = c_n\, e^{-jn\omega_0 \times 3 \times 10^{-9}} = c_n\, e^{-j2\pi n(3/10)} = c_n\, e^{-j0.6\pi n}$. (b) No --- $|e^{-j\theta}| = 1$ for all θ, so magnitude of every harmonic is unchanged. *(chap3.pdf, p.20)*

1.5.4 Chapter 4 Answers

A4.1 *(chap4.pdf, p.3)* (1) Time delay: $T_D = L/v$; (2) reflections from impedance mismatch.

A4.2 *(chap4.pdf, p.82)* Skin effect --- current concentrates in a thin annulus of depth $\delta = 1/\sqrt{\pi f \mu \sigma}$, reducing the effective cross-section.

A4.3 *(chap4.pdf, pp.4--5)* Coaxial: fields entirely inside the homogeneous dielectric fill $\to lc$ is constant, v is constant $\to$ non-dispersive. Microstrip: field partially in air and partially in dielectric $\to$ effective permittivity varies with frequency $\to$ dispersive.

A4.4 *(chap4.pdf, p.12)* $lc = \mu\varepsilon$; therefore $v = 1/\sqrt{lc} = 1/\sqrt{\mu\varepsilon}$ is determined entirely by the medium properties.

A4.5 Open circuit ($Z_L = \infty$): $\Gamma_V = +1$ (voltage doubles), $\Gamma_I = -1$ (current reverses --- it must be zero at the open end). Short circuit ($Z_L = 0$): $\Gamma_V = -1$ (voltage = 0), $\Gamma_I = +1$.

A4.6 *(chap4.pdf, pp.54--55)* Series: $Z_S + R = Z_C$; load voltage = full V_0; no steady-state power loss. Parallel: $R = Z_C$ (for high-Z load); load voltage < source; resistor draws current and dissipates power.

A4.7 *(chap4.pdf, p.59)* (1) $\tau_r > 10\,T_D$ (rise time criterion); (2) $L < \lambda/10$ (electrical-length criterion).

A4.8 *(chap4.pdf, p.87)* Dielectric: $g = \omega c \tan\theta \propto f$ (20 dB/decade). Skin effect: $r \propto \sqrt{f}$ (10 dB/decade). At very high frequencies, dielectric loss overtakes conductor loss.

A4.9 *(chap4.pdf, pp.92--94)* Low-loss valid above ˜5 MHz; β dispersion disappears above ˜20 MHz; α dispersion from skin effect persists above 20 MHz (no, it never completely disappears in the skin-effect region).

A4.10 $l \approx (\mu_0/\pi)\ln(d/a) = (4\pi \times 10^{-7}/\pi)\ln(20/1) = 4 \times 10^{-7} \times 2.996 = 1.198\,\mu$H/m; $c \approx \pi\varepsilon_0/\ln(d/a) = \pi \times 8.854 \times 10^{-12}/2.996 = 9.28$ pF/m. *(chap4.pdf, pp.19--22)*

A4.11 (a) $l = (\mu_0/2\pi)\ln(5) = (2\times10^{-7})\times1.609 = 321.8$ nH/m; (b) $c = 2\pi\varepsilon_0\varepsilon_r/\ln(5) = 2\pi\times8.854\times10^{-12}\times2.25/1.609 = 77.7$ pF/m; (c) $Z_C = \sqrt{l/c} = \sqrt{321.8\times10^{-9}/77.7\times10^{-12}} = \sqrt{4141} = 64.3\,\Omega$; (d) $v = 1/\sqrt{lc} = 1/\sqrt{321.8\times10^{-9}\times77.7\times10^{-12}} = 2\times10^8$ m/s $= v_0/\sqrt{2.25} = v_0/1.5$. *(chap4.pdf, pp.25--27)*

A4.12 (a) $T_D = 0.3/(2\times10^8) = 1.5$ ns; (b) $V^+_{\text{init}} = 10\times75/(75+75) = 5$ V; (c) $\Gamma_L = (300-75)/(300+75) = 225/375 = 0.6$; $\Gamma_S = (75-75)/(75+75) = 0$; (d) $V_L(T_D) = 5\times(1+0.6) = 8$ V; since $\Gamma_S = 0$, no re-reflection $\rightarrow V_L(3T_D) = 8$ V; $V_L(\infty) = 10\times300/375 = 8$ V (consistent). *(chap4.pdf, pp.34--37)*

A4.13 $\tau_r/T_D = 2$ ns$/0.3$ ns $= 6.7 < 10$. The criterion is NOT met. The line will significantly distort the waveform (overshoot expected). *(chap4.pdf, p.59)*

A4.14 $\Gamma_L = (150+j50-50)/(150+j50+50) = (100+j50)/(200+j50)$; $|100+j50| = \sqrt{100^2+50^2} = 111.8$; $|200+j50| = \sqrt{200^2+50^2} = 206.2$; $|\Gamma_L| = 111.8/206.2 = 0.542$. VSWR $= (1+0.542)/(1-0.542) = 1.542/0.458 = 3.37$. Power to load $= 1-|\Gamma_L|^2 = 1-0.294 = 0.706 = 70.6\%$. *(chap4.pdf, pp.77--78)*

A4.15 (a) $\delta = 1/\sqrt{\pi\times10^8\times4\pi\times10^{-7}\times5.8\times10^7} = 1/\sqrt{\pi\times4\pi\times10^8\times5.8} = 1/\sqrt{7.27\times10^9} = 6.63\times10^{-6}$ m $= 6.63\,\mu$m; (b) $r_w = 1$ mm $= 1000\,\mu$m $\gg 6.63\,\mu$m $\rightarrow$ yes, HF formula applies; (c) $r_{hf} = 1/(2\pi r_w\sigma\delta) = 1/(2\pi\times10^{-3}\times5.8\times10^7\times6.63\times10^{-6}) = 1/(2\pi\times384.5) = 4.14\times10^{-4}\,\Omega$/m $= 0.414$ mΩ/m. *(chap4.pdf, pp.82--83)*